Longitudinal Metagenetic Analysis of Microbial Transmission Dynamics and Antibiotic Resistance Gene Dissemination between Hand and Residential Surface

BIO303 Final Year Project

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2. Experiment Design and Objectives
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Background

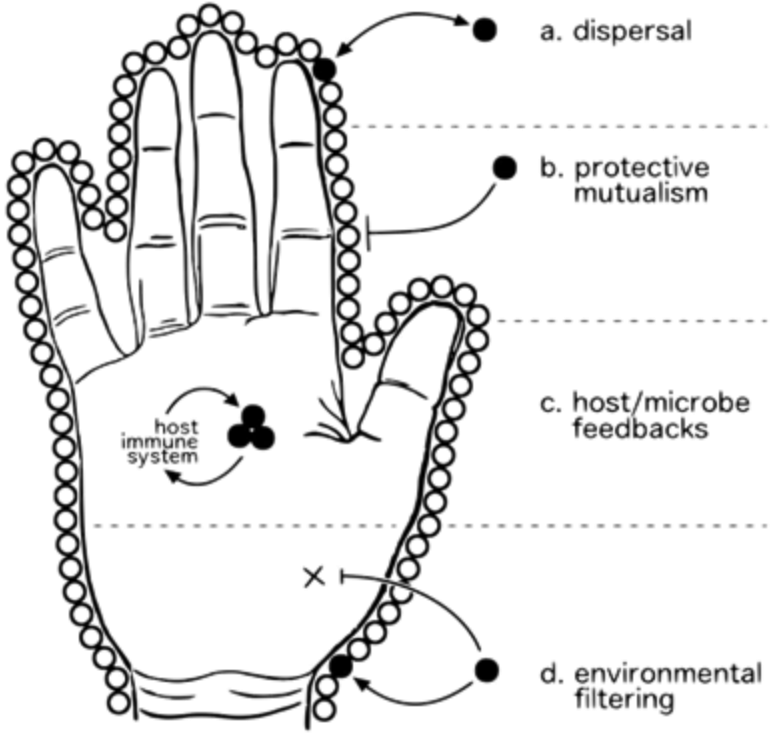


Figure 1. The concept of hygiene and the hand microbiome. (Vandegrift, 2017)

Figure 2. Proposed model for the hand as a critical vector in microbiome dynamics. (Edmonds-Wilson, Nurinova et al. 2015)

Background

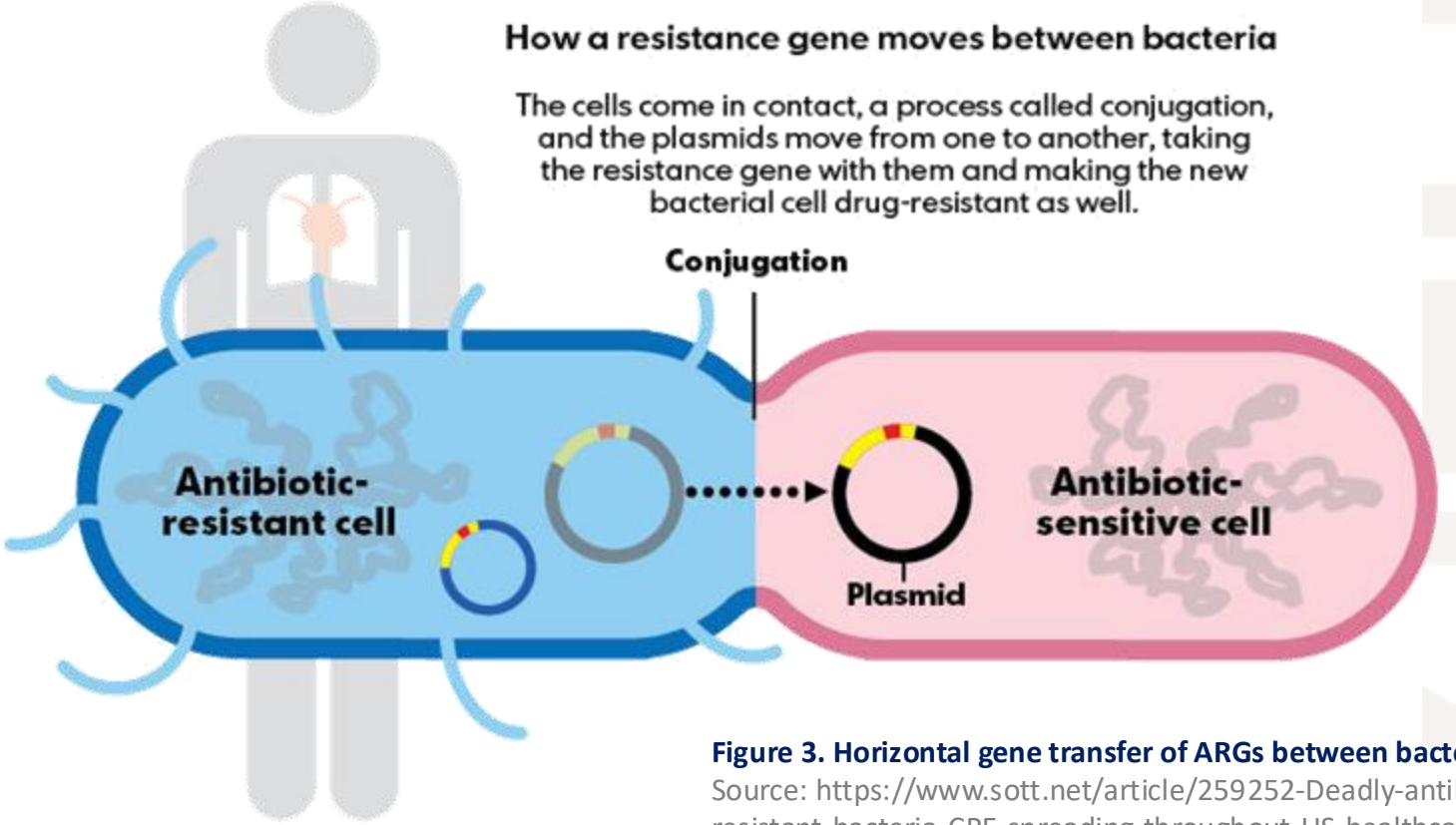


Figure 3. Horizontal gene transfer of ARGs between bacteria.
Source: https://www.sott.net/article/259252-Deadly-antibiotic-resistant-bacteria-CRE-spreading-throughout-US-healthcare-systems

Experiment design

Occupant 1

Residence 1

Day 1 → Day 2 → … → Day 10

Occupant 2

Residence 2

Day 1 → Day 2 → … → Day 10

Occupant 3

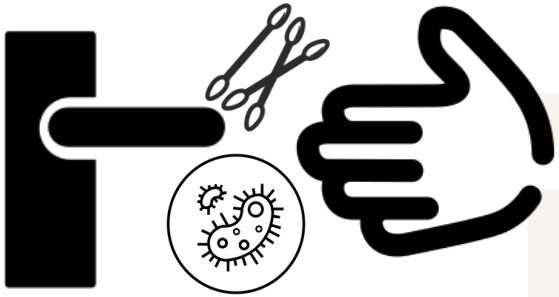
Residence 3

Day 1 → Day 2 → … → Day 10

Occupant 4

Residence 4

Day 1 → Day 2 → … → Day 10



Sampling at

- 9 a.m.
- 6 p.m.

for successive 10 days

Community level (low resolution)

Objectives

Occupant 3

Residence 3

Research goals:

- Short-term dynamics and assembly pattern of hand and residential surface microbiome
- Transmission pattern of microbial taxa and ARGs during hand-surface touch

Genome level (high resolution)

Methods

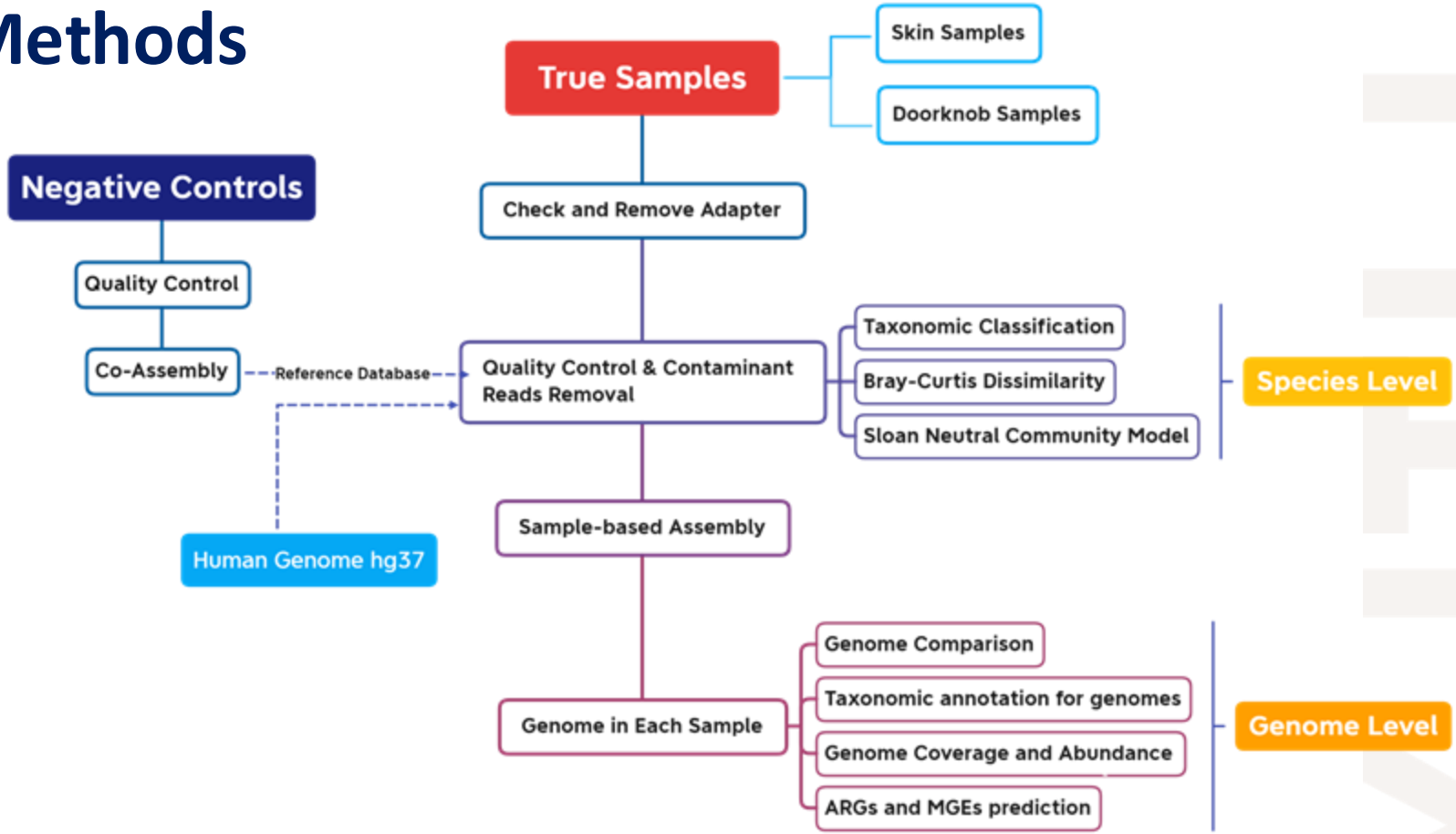
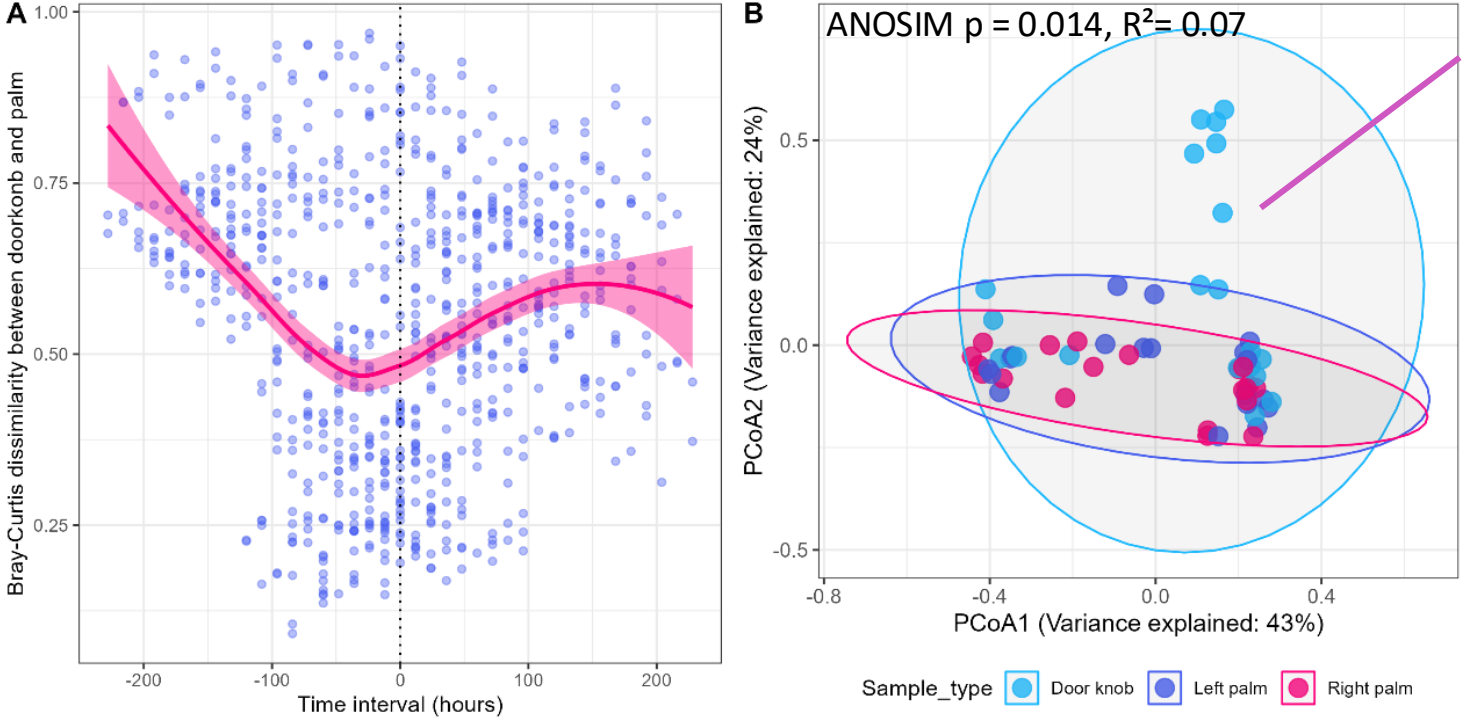


Figure 4. Workflow of Data Analysis on negative controls and Skin Samples

Results

Bray-Curtis dissimilarity between palms and doorknob



low similarity between doorknob and palms

Figure 5. Bray-Curtis dissimilarity and Principal coordinate analysis (PCoA) between doorknob and palms

Results

Taxonomy Classification

Top 3 most abundant taxa: *Gordonia bronchialis*, *Janibacter indicus*, and *Micrococcus luteus*

Belonged to the Micrococcales and Mycobacteriales orders

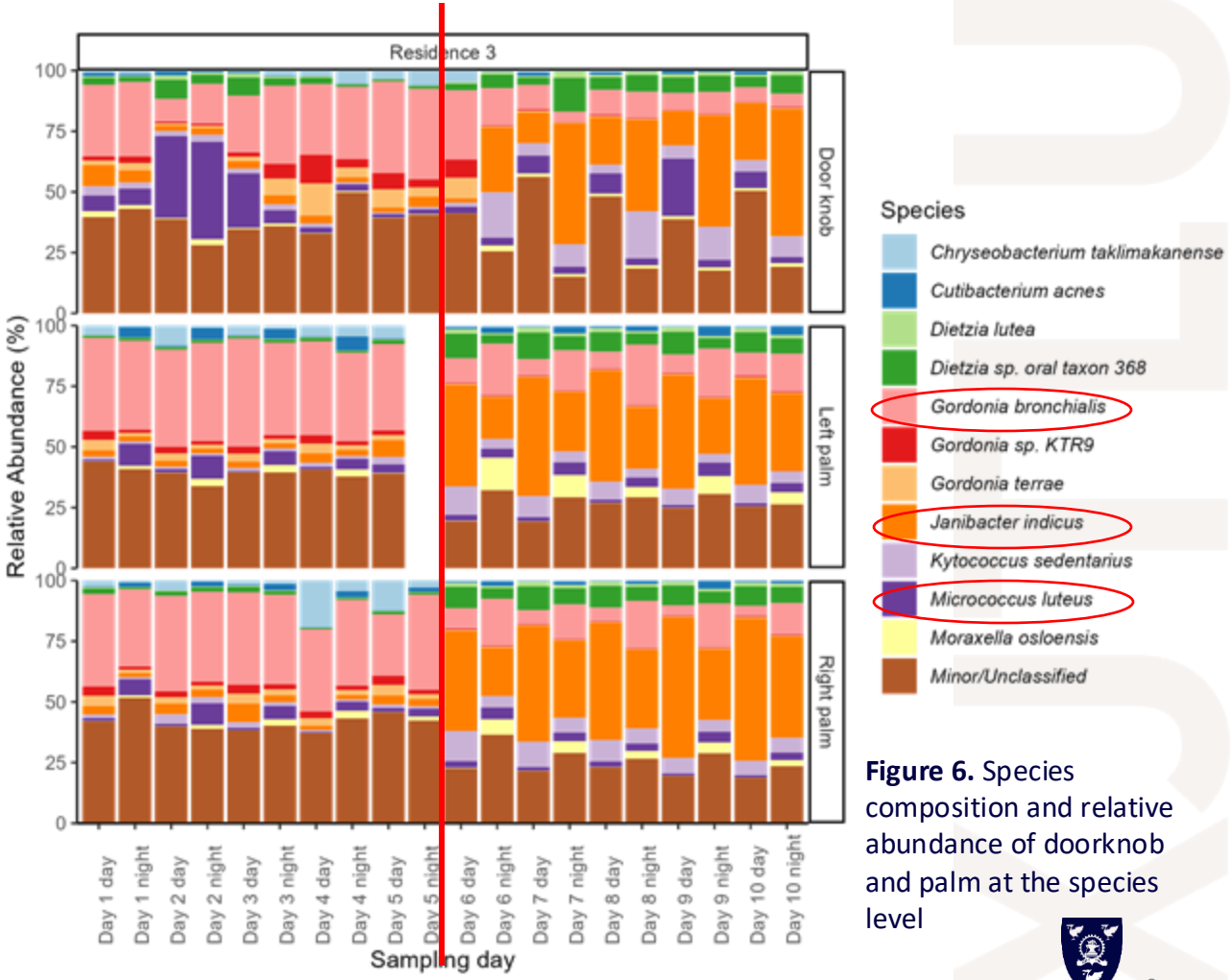


Figure 6. Species composition and relative abundance of doorknob and palm at the species level

Results

Taxonomy Classification

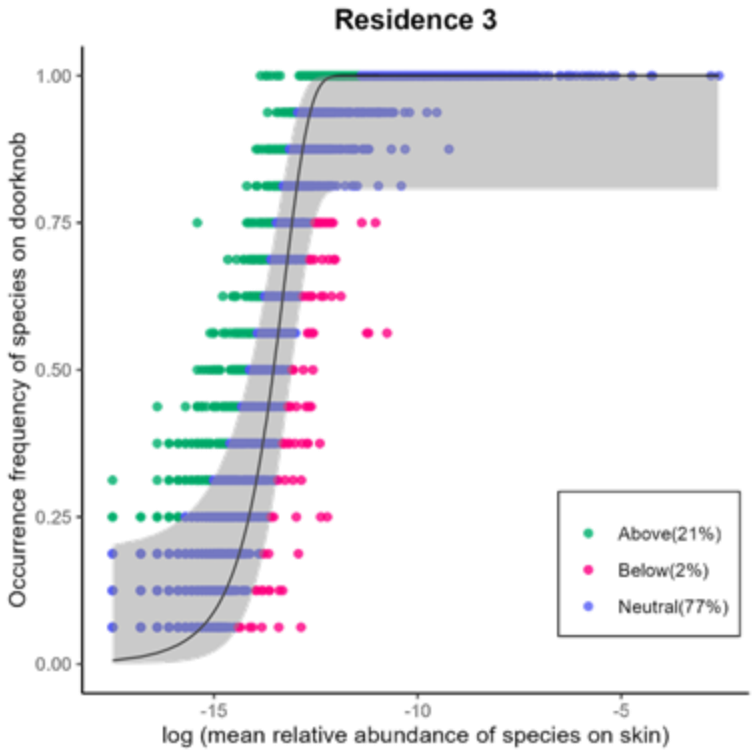


Figure 7. Sloan neutral community model (NCM) for the dispersal of microbial taxa from hands to doorknob

- *Mycobacterium tuberculosis*
- *Staphylococcus pettenkoferi*
- *Staphylococcus cohnii*
- …

Partition	Number of Species	Number of Pathogen
Above (5.75%)	818	47
Neutral (6.10%)	2966	181
Below (16.48%)	91	15

Table 1. The proportion of pathogens identified on different partitions

- Above: strong dispersal capacity
- Neutral: normal dispersal capacity
- Below: deficient dispersal capacity

Results

Microbial transmission pattern at the genome level

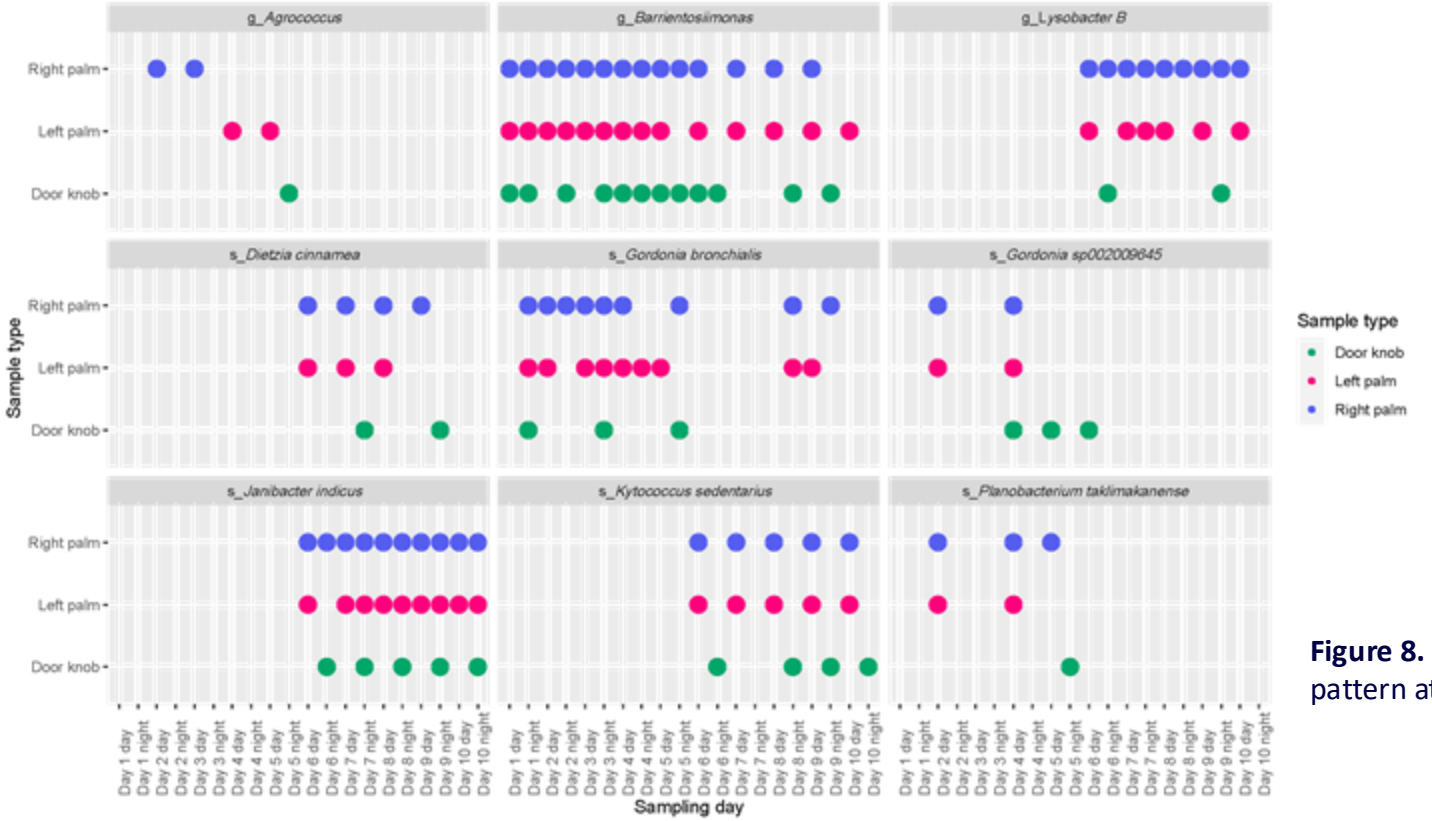


Figure 8. Microbial transmission pattern at the genome level

Results

Microbial transmission pattern at the genome level

Figure 9. Coverage of rMAGs in each sample

Results

Identification of antibiotic protein families

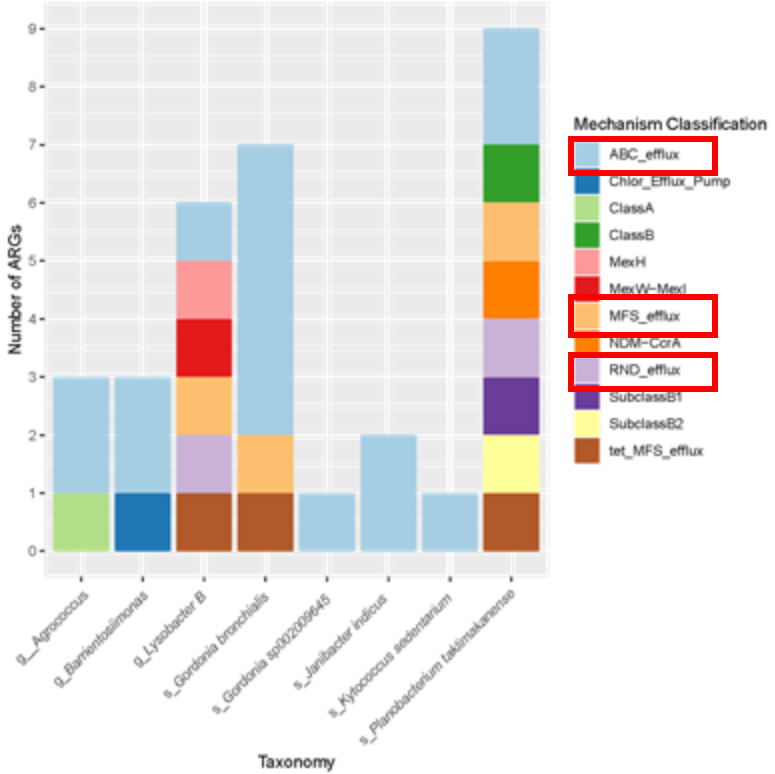


Figure 10. Antibiotic resistance protein families of rMAGs

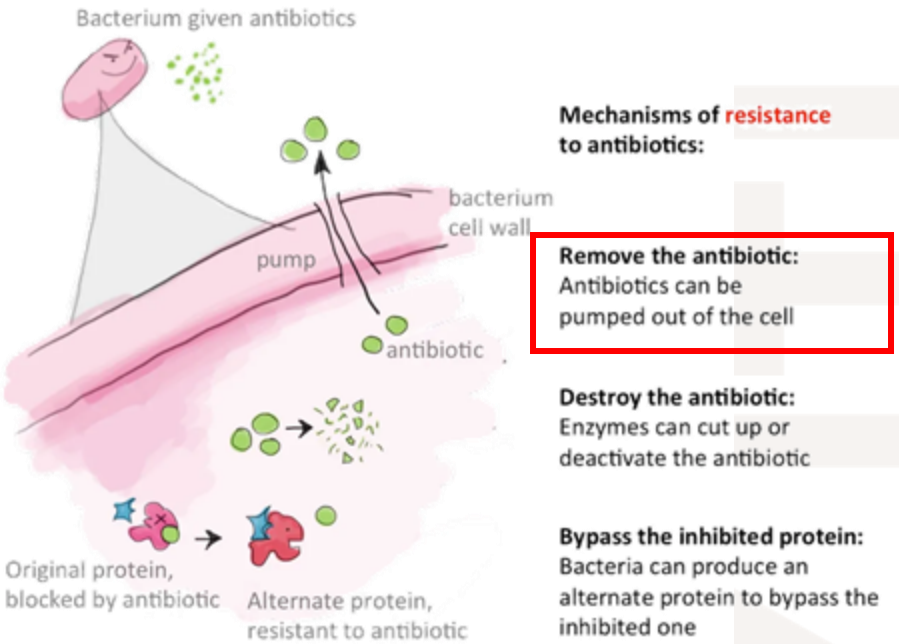


Figure 11. Mechanisms of resistance to antibiotic
Source:https://sitn.hms.harvard.edu/flash/2016/antibiotic-resistance-old-genes-new-problems/

Results

Mobile Genetic Elements and Antibiotic Resistance Genes Identification

Taxonomy	nIS	%Genome	bps4IS	dnaLen
s_*Planobacterium taklimakanense*	20	1.3	33659	2596718
s_*Gordonia bronchialis*	30	0.8	36098	4530003
g_*Agrococcus*	8	0.77	14146	1841737
s_*Janibacter indicus*	14	0.59	17087	2897137
g_*Barrientosiimonas*	11	0.44	15718	3543856
s_*Kytococcus sedentarius*	8	0.35	6887	1995664
g_*Lysobacter B*	6	0.26	4624	1812744
s_*Gordonia sp002009645*	7	0.19	7467	3872203

Table 2. Insertion Sequence (IS) detected in contigs

*nIS:the number of IS copies; %Genome: the percentage of the genome spanned by IS sequences; bps4IS: length of sequence segment spanned by the IS sheet in the sequence jdnaLen: total nucleic acid length of the inserted column

Results

Mobile Genetic Elements and Antibiotic Resistance Genes Identification

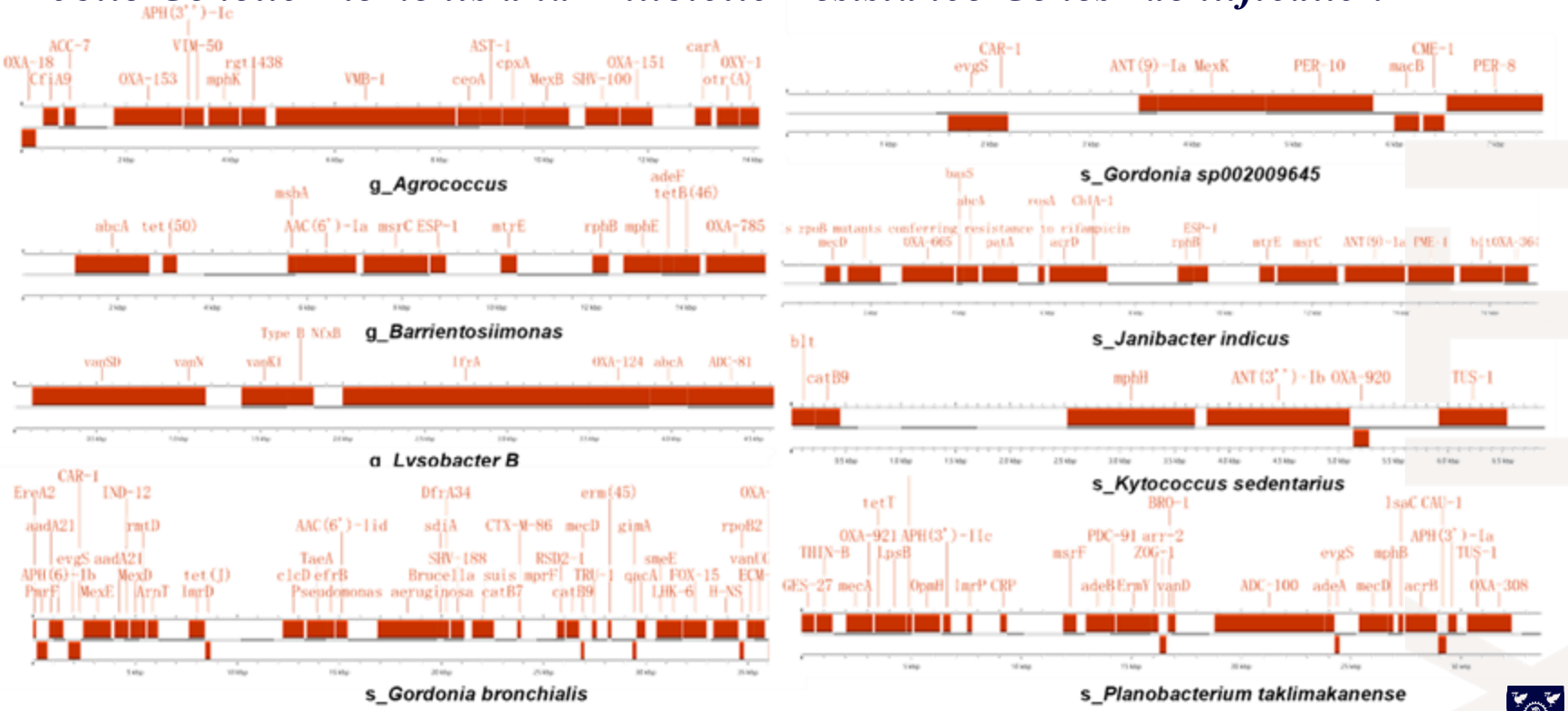


Figure 12. ARGs in insertion sequence identified in rMAGs

Discussion

- Transmission of microbial taxa between humans and residential surfaces
- Investigating the diffusion of ARGs in indoor environments during microbiota exchange

Limitation

- Insufficient subject characteristics
- Small sample size
- Low quality sequences

Conclusion

Significance

- Reduce the adverse effects of disease
- Enhance health and well-being

In the Future…

- Bidirectional microbial transmission patterns and their implications for human health
- Effective strategies to control the spread of antibiotic resistance in indoor environments

Source: https://blog.amopportunities.org/2021/02/11/heres-what-to-expect-during-u-s-medical-residency/

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Thank you

Any suggestions or criticism will be appreciated.